

Paracetamol (acetaminophen) poisoning: The early years.

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Paracetamol (acetaminophen) was marketed in the 1950s as a nonprescription analgesic/antipyretic without any preclinical toxicity studies. It became used increasingly for self-poisoning, particularly in the UK and was belatedly found to cause acute liver damage, which could be fatal. Management of poisoned patients was difficult as maximum abnormalities of liver function were delayed for 3 days or more after an overdose. There was no treatment and the mechanism of hepatotoxicity was not known. The paracetamol half-life was prolonged with liver damage occurring when it exceeded 4 h and the Rumack-Matthew nomogram was an important advance that allowed stratification of patients into separate zones of risk. It is used to guide prognosis and treatment and its predictive value could be increased by combining it with the paracetamol half-life. The problems of a sheep farmer in Australia in the early 1970s led to the discovery of the mechanism of paracetamol hepatotoxicity, and the first effective treatment of overdosage with intravenous (IV) cysteamine. This had unpleasant side effects and administration was difficult. N-acetylcysteine soon became the treatment of choice for paracetamol overdose and given early it was very effective when administered either IV or orally. N-acetylcysteine could cause anaphylactoid reactions, particularly early during IV administration when the concentrations were highest. Simpler and shorter regimes with slower initial rates of infusion have now been introduced with a reduced incidence of these adverse effects. In addition, there has been a move to use larger doses of N-acetylcysteine given over longer periods for patients who are more severely poisoned and those with risk factors. There has been much interest recently in the search for novel biomarkers such as microRNAs, procalcitonin and cyclophilin that promise to have greater specificity and sensitivity than transaminases. Paracetamol-protein adducts predict hepatotoxicity and are specific biomarkers of toxic paracetamol metabolite exposure. Another approach would be measurement of the plasma levels of cysteine and inorganic sulfate. It is 50 years since the first effective treatment for paracetamol poisoning and, apart from liver transplantation, there is still no effective treatment for patients who present late.

Comparison of Three-Bag Method Acetylcysteine Versus Two-Bag Method Acetylcysteine for the Treatment of Acetaminophen Toxicity: An Updated Systematic Review and Meta-Analysis.

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Acetaminophen is generally considered safe when used according to the recommended guidelines. Consumption in excessive doses can lead to severe liver damage and, in critical cases, may even result in death. To reduce the effects of acetaminophen overdose, N-acetylcysteine (NAC) has been established as the preferred intervention to prevent liver damage. The purpose of this updated systematic review and meta-analysis is to evaluate the potential benefits of a two-bag N-acetylcysteine (NAC) dosing regimen compared to the traditional three-bag protocol in the treatment of acetaminophen-induced liver toxicity. This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. The research team utilized the PubMed and Cochrane databases to perform a thorough and comprehensive search of the relevant literature from the inception of these databases up until January 2024. Nine studies were included. The overall use of two-bag NAC was associated with lower anaphylactic reactions and gastrointestinal symptoms compared to the three-bag method. The rate of liver toxicity resolution was the same between the two treatment groups. The two-bag NAC regimen can be considered a safe and effective method for managing acetaminophen

toxicity.

[Side-effects of N-acetylcysteine treatment in patients with paracetamol poisoning].

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Treatment of paracetamol intoxication with N-acetylcysteine (NAC) is standard in Denmark. NAC is considered safe with relatively few side effects. It is recommended that all patients be treated irrespective of paracetamol dose or time from intoxication to treatment start. Consequently a higher number of patients will be treated with NAC than with previous regimens based on plasma concentrations of paracetamol. In this retrospective study we evaluated the incidence of side effects of NAC in 310 patients admitted to the Department of Hepatology, Rigshospitalet, Copenhagen, over a four-year period (1.1.1994-31.12.1997). Twenty-six (8.4%) patients developed side effects. Side effects were anaphylactoid, mainly from skin (25 rash, pruritus or flushing), in rare cases more serious (four bronchospasm, three angioedema, one hypotension). None were life-threatening and all patients received the full course of NAC. In all cases the recommended treatment with antihistamine or steroids against adverse effects was administered. We conclude that treatment with NAC is safe. Accordingly we find no reason to change the recommendation for treatment of paracetamol intoxication in Denmark.

Non-narcotic analgesics. Problems of overdosage.

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The first cases of fulminant hepatic failure due to paracetamol poisoning were reported in 1966, and in the United Kingdom this condition is now responsible for more cases of acute hepatic failure than any other cause. Adults account for the majority of serious and fatal cases of paracetamol poisoning and it is extremely rare for young children to ingest sufficient paracetamol to cause more than minimal liver damage. A single measurement of the plasma paracetamol concentration is an accurate predictor of liver damage provided that it is taken not earlier than 4 hours after ingestion of the overdose. Peak disturbance of liver function occurs 2 to 4 days after the overdose, often accompanied by mild jaundice, after which recovery is usually rapid and complete. In a few patients, fulminant hepatic failure, manifested by increasing jaundice and encephalopathy, may develop by the third to fifth day. Acute renal failure may complicate paracetamol poisoning, often in the context of severe liver damage. Renal failure, which is often non-oliguric, typically becomes apparent 24 to 72 hours after overdosage. The treatment of paracetamol intoxication should include gastric lavage, which has been shown to be of value for up to 6 hours after ingestion of a paracetamol overdose. Further general treatment may include parenteral fluid replacement and a prophylactic infusion of dextrose (5-10%) in patients at risk of hepatic failure. Specific protective agents in those patients at risk of paracetamol-induced liver damage include N-acetylcysteine and methionine which are most effective if given within 8 to 10 hours of ingestion of the overdose. Hepatic and renal failure should be managed conventionally. In recent years in the United Kingdom there has been a gradual decline in the number of hospital admissions and the number of deaths from aspirin poisoning. Salicylates in overdose directly stimulate the respiratory centre and so cause a respiratory alkalosis. Metabolic acidosis occurs in severe poisoning because of impairment of the oxidative metabolism of energy substrates. At very high salicylate concentrations respiratory depression may occur, possibly associated with neuroglycopenia, adding respiratory acidosis to the worsening metabolic acidosis. In addition to a mixed acid-base disturbance, hypokalaemia and hypoglycaemia may

be present. Nausea and vomiting increase the fluid deficit. If dehydration is sufficiently severe, decreasing cardiac output may hasten development of lactic acidosis and acute renal failure.(ABSTRACT TRUNCATED AT 400 WORDS)

Large paracetamol overdose-Higher dose acetylcysteine is required.

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Paracetamol poisoning continues to be a worldwide problem and, despite the availability of an effective antidote, acetylcysteine (NAC), the optimal way to use this antidote, particularly following very large doses of paracetamol, has not been established. Recent case series have shown an increased toxicity from high doses of paracetamol, even in those receiving prompt NAC therapy, particularly in patients above the 300 mg/L nomogram treatment line. Clinical trial evidence supporting shorter NAC dosing now allows the possibility for intensifying treatment without the risk of very high rates of ADRs. New biomarkers also show the possibility of early identification of patients at risk of liver injury who might also benefit from increased intensity treatment. This article discusses these data and proposes a logical therapy for increasing NAC dosing which now requires clinical trial testing.

Predicting the requirement for N-acetylcysteine in paracetamol poisoning from reported dose.

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There is contention over whether reported dose correlates with toxicity in paracetamol poisoning and risk assessment is currently based on serum paracetamol concentration compared to a nomogram, irrespective of reported dose. Objective. To determine if reported dose predicts the need for N-acetylcysteine (NAC). Data were taken from paracetamol overdoses presenting to a tertiary toxicology service. Age, sex, reported dose, ingestion time, timed paracetamol concentrations between 4 and 16 h, hepatotoxicity (peak alanine transaminase > 1000 U/L) and treatment (single dose-activated charcoal [SDAC] and NAC) were analysed. Data were analysed within a repeated measures logistic regression framework using NONMEM (ver 7.2). The primary outcome was administration of NAC, which was determined based on a serum paracetamol concentration greater than the nomogram line. There were 1571 admissions in 1303 patients, with a median age of 27 years (12-96 years) and 1140 (73%) were females. The median dose was 10 g (1-100 g). The paracetamol concentration was above the nomogram line in 337 of 1571 (22%) patients. Patients presenting later (first paracetamol concentration between 7 and 16 h post-overdose) compared to those presenting earlier (4-7 h post-overdose) were more likely to have hepatotoxicity (5.5% vs. 0.4%; $p < 0.0001$), have a toxic paracetamol concentration (34% vs. 18%; $p < 0.0001$) and receive NAC (48% vs. 23%; $p < 0.0001$). SDAC reduced the probability of the paracetamol concentration being above the nomogram. Based on SDAC not being administered there was a 5% probability of requiring NAC at a dose of 6-9 g, a 10% chance of requiring NAC at a dose of 13-16 g, a 50% chance of requiring NAC at a dose of 30-34 g and a 90% chance for needing NAC at 48-50 g. Reported dose was a good predictor of a toxic paracetamol concentration and SDAC reduced the probability of the concentration being above the nomogram. These predictions may assist in determining which patients could be started on NAC immediately.

[Improved and harmonised laboratory analysis of paracetamol provides safer assessment of poisoning cases].

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Toxicological analysis is an important part of the acute treatment of various intoxications. Rapid laboratory responses are important for the patient to be assessed and treated correctly, and also to exclude poisoning and thus avoid unjustified and costly overtreatment. In Sweden, paracetamol (acetaminophen) is one of the most common pharmaceuticals in drug poisoning. Paracetamol overdose can cause severe liver damage unless treated early with the antidote acetylcysteine. A nation-wide initiative for improved laboratory measurement of paracetamol in plasma/serum samples has resulted in a marked reduction in the inter-laboratory coefficient of variation to generally below 10%. The introduction of a harmonized national reporting range for plasma/serum paracetamol covering at least 50-5 000 $\mu\text{mol/l}$ was also recommended. This initiative will hopefully contribute to better healthcare from both a patient and health resource perspective in cases of paracetamol poisoning.

[The influence of the dose, time since ingestion and concentration of the xenobiotic on the clinical state and severity of liver damage with patients intoxicated with paracetamol].

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The aim of this study was to determine relations between the clinical state and the severity of liver damage comparing to the amount of ingested paracetamol, time since ingestion and serum concentration of paracetamol with patients after acute intoxication with this drug. A retrospective analysis of medical records of 95 patients hospitalized in the Toxicology Department in Kraków or treated in the Toxicological Admission Room in 2002-2004 years because of acute paracetamol intoxication was performed. The general clinical state connected with intoxication and the severity of liver damage were determined using the Poisoning Severity Score (PSS). The level of consciousness was determined with the Glasgow Coma Scale and Matthew's scale. The serum concentration of paracetamol during admission to the hospital, levels of biochemical markers of liver damage (AST, ALT), bilirubin and INR index were also performed. We found a statistically significant positive correlation between the ingested dose of paracetamol comparing to the gravity of poisoning, the severity of liver damage, levels of aminotransferases and bilirubin. A positive correlation between time since ingestion of paracetamol to hospitalization and the gravity of poisoning according to PSS scale was also statistically significant. A paracetamol concentration measured during admission to the hospital had no influence on neither the clinical state of patient nor the severity of liver damage. We conclude that the therapeutical approach should consider possibly shortening of the time since drug ingestion to hospitalization and institution of specific treatment (N-acetylcysteine) as well as minimalization of the paracetamol dose, which could be absorbed, by different methods of elimination from the GI tract (eg., gastric lavage, activated charcoal laxatives).

Intox, detox, antidotes - Evidence based diagnosis and treatment of acute intoxications.

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Aim of this review is to describe the role of clinical toxicology in the context of acute medicine. A special focus is put on antidotes and important aspects of diagnosis and therapy of acute intoxications. The data of the annual report of GIZ-Nord Poisons Centre is analyzed concerning the following aspects: what intoxications are relevant in acute medicine, are there special aspects in therapy, e.g. antidotes, and what antidotes are relevant? More over intoxication-related fatalities are analyzed. In 2015 the poisons centre was consulted in 33,000 cases of acute intoxications. The most important groups are drugs (e.g. antidepressants, beta blockers and calcium channel blockers), chemical products (e.g. products containing surfactant, corrosive substances and toxic alcohols like methanol), plants and recreational drugs. Intoxications are relevant in acute medicine. Some substances can cause fatal intoxications. Important antidotes are naloxone for opioids, acetylcysteine for paracetamol, fomepizole and ethanol for toxic alcohols and diazepam for intoxications caused by chloroquine.

The Mechanism and Inflammatory Markers Involved in the Potential Use of N-acetylcysteine in Chronic Pain Management.

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N-acetylcysteine (NAC) has established use as an antidote for acetaminophen overdose and treatment for pulmonary conditions and nephropathy. It plays a role in regulating oxidative stress and interacting with various cytokines including IL-1 β , TNF α , IL-8, IL-6, IL-10, and NF- κ B p65. The overexpression of reactive oxygen species (ROS) is believed to contribute to chronic pain states by inducing inflammation and accelerating disease progression, favoring pain persistence in neuropathic and musculoskeletal pain conditions. Through a comprehensive review, we aim to explore the mechanisms and inflammatory pathways through which NAC may manage neuropathic and musculoskeletal pain. Evidence suggests NAC can attenuate neuropathic and musculoskeletal pain through mechanisms such as inhibiting matrix metalloproteinases (MMPs), reducing reactive oxygen species (ROS), and enhancing glutamate transport. Additionally, NAC may synergize with opioids and other pain medications, potentially reducing opioid consumption and enhancing overall pain management. Further research is needed to fully elucidate its therapeutic potential and optimize its use in pain management. As an adjuvant therapy, NAC shows potential for chronic pain management, offering significant benefits for public health.
